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How to remove the seasonal atmospheric effect: An application on a surface volcanic temperature at Nisyros active volcano, Hellenic Volcanic Arc

A. Marsellos, Konstantinos Hantzis, K. Tsakiri •
Published in Journal of Volcanology and… 1 December 2020 • Environmental Science, Geology

Abstract Surface temperature at a volcanic system is the result of a mixture of two heat sources originating one from above and another from below the ground surface. The atmosphere significantly affects the surface's temperature of the volcano, and removal of the atmospheric effect is required for meaningful interpretations prior to any characterization of anomalous volcanic activity. Time series of meteorological data were acquired from the closest weather station to Nisyros volcano, and… Expand

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Sophie Pailot-Bonnétat, Andrew J. L. Harris, +6 authors, Michael Ramsey • Environmental Science, Geology •

Bulletin of Volcanology • 2025

At active volcanic hydrothermal systems, enhanced heat flow results in surface heating and elevated heat flux densities. We consider heat transfer models for two types of system: (1) “dry” (soil… Expand

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Marsellos Ae, Tsakiri Kg, S. Kapetanakis, K. Kyriakopoulos · Environmental Science, Geology · 2017

Nisyros volcano has shown a series of volcanic and seismo-tectonic events including recent volcanic phreatic eruptions intruding an impermeable layer near the surface. Geological and atmospheric… Expand

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A. Rinaldi, J. Vandemeulebrouck, M. Todesco, F. Viveiros · Environmental Science, Geology · 2012

[1] Diffuse degassing through the soil is commonly observed in volcanic areas and monitoring of carbon dioxide flux at the surface can provide a safe and effective way to infer the state of activity… Expand

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S. Oliveira, F. Viveiros, Catarina Silva, J. Pacheco · Environmental Science, Geology · Frontiers in Earth Science · 2018

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Geology, Physics · Solid Earth ·

F. Dinger S. Bredemeyer S. Arellano N. Bobrowski U. Platt T. Wagner

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D. Granieri G. Chiodini W. Marzocchi R. Avino Environmental Science, Geology · 2003

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G. Chiodini S. Caliro +5 authors C. Minopoli Environmental Science, Geology · 2010

[1] Long-duration time series of the chemical composition of fumaroles and of soil CO2 flux reveal that important variations in the activity of the Solfatara fumarolic field, the most important… Expand

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Y. Taran D. Rouwet S. Inguaggiato A. Aiuppa Geology, Environmental Science · 2008

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Y. Shimoike K. Notsu Environmental Science, Geology · 2000

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T. Girona C. Huber C. Caudron Geology, Environmental Science · Scientific Reports · 2018

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